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Ex.No.7: Analog Beamforming Analysis Using MATLAB

Objective 1:

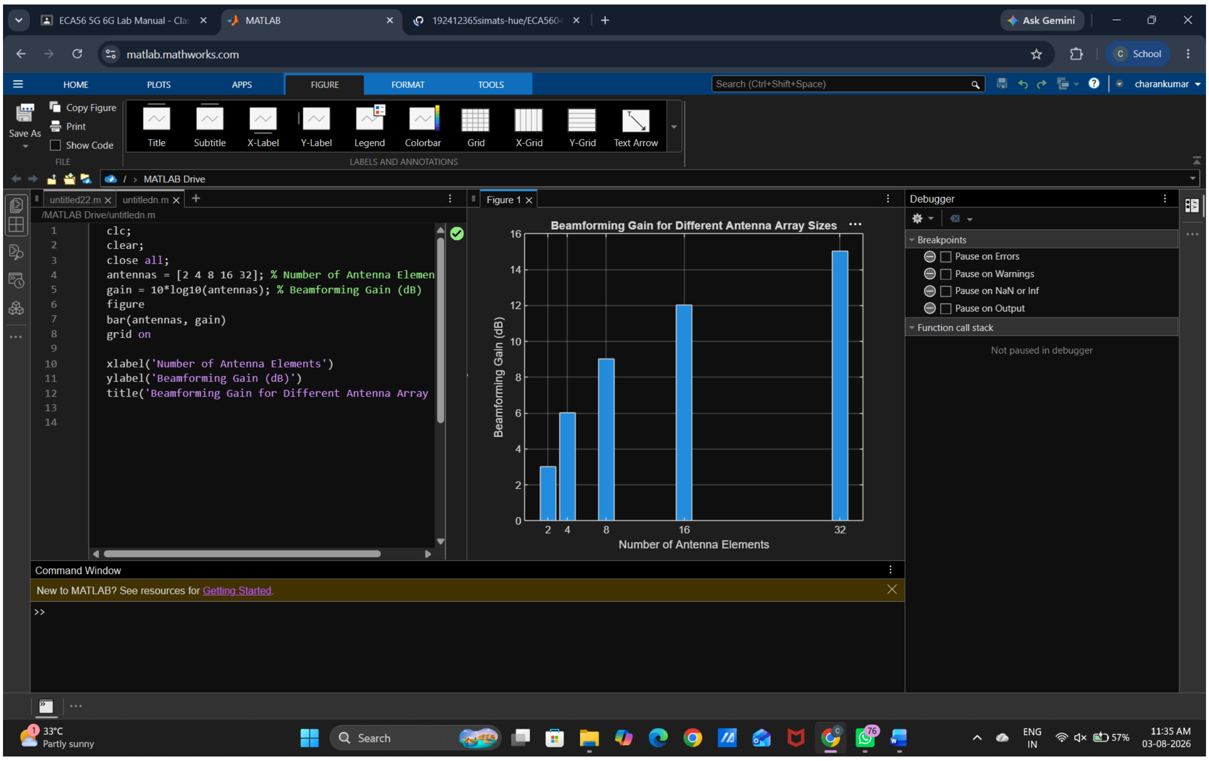
To analyze the beamforming gain achieved by varying the number of antenna elements in an analog beamforming system using MATLAB.

Program (Matlab Code)

```
clc;
clear;
close all;
antennas = [2 4 8 16 32]; % Number of Antenna Elements
gain = 10*log10(antennas); % Beamforming Gain (dB)
figure
bar(antennas, gain)
grid on

xlabel('Number of Antenna Elements')
ylabel('Beamforming Gain (dB)')
title('Beamforming Gain for Different Antenna Array Sizes')
```

Output:



Objective 2:

To compare the beam width of an antenna array for different beamforming configurations using MATLAB.

Matlab Code

clc;

clear;

close all;

antennas = [2 4 8 16 32]; % Number of Antenna Elements

beamwidth = [90 45 22.5 11.25 5.625]; % Beam Width (degrees)

figure

plot(antennas, beamwidth,'o-','LineWidth',2)

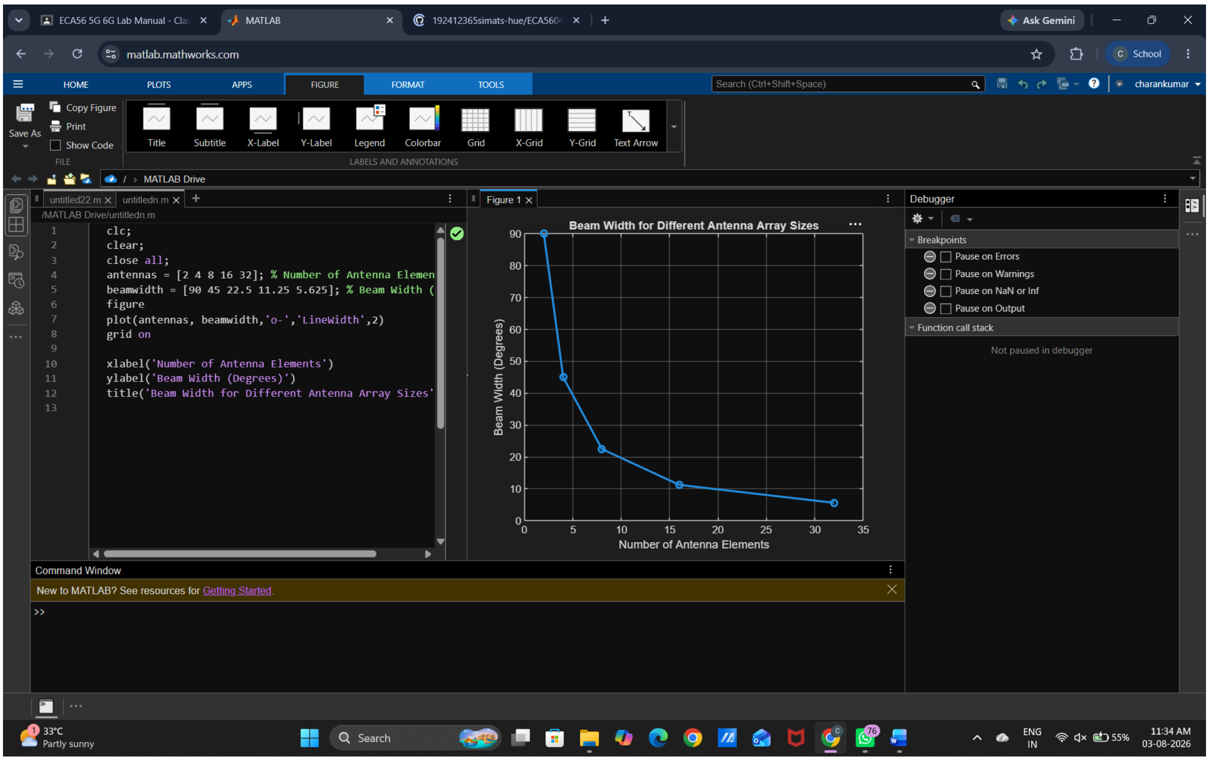
grid on

xlabel('Number of Antenna Elements')

ylabel('Beam Width (Degrees)')

title('Beam Width for Different Antenna Array Sizes')

Output



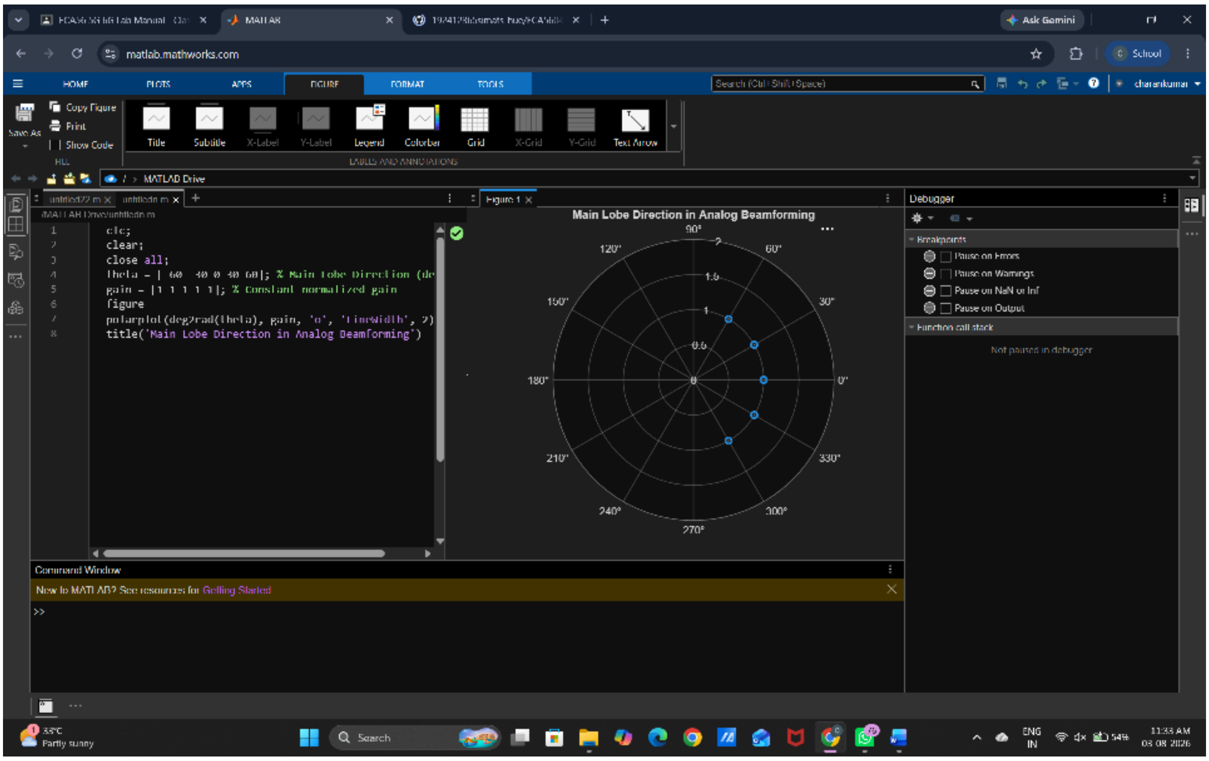
Objective 3:

To analyze the main lobe direction by steering the antenna array beam to different angles MATLAB.

Matlab Code

```
clc;
clear;
close all;
theta = [-60 -30 0 30 60]; % Main Lobe Direction (degrees)
gain = [1 1 1 1 1]; % Constant normalized gain
figure
polarplot(deg2rad(theta), gain, 'o', 'LineWidth', 2)
title('Main Lobe Direction in Analog Beamforming')
```

Output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clc;
clear;
close all;
theta = [-60 -30 0 30 60]; % Main Lobe Direction (de
gain = [1 1 1 1 1]; % Constant normalized gain
figure
polarplot(deg2rad(theta), gain, 'o', 'LineWidth', 2)
title('Main Lobe Direction in Analog Beamforming')
Figure 1
Main Lobe Direction in Analog Beamforming
90°
60°
30°
0°
330°
300°
270°
240°
210°
180°
150°
120°
2
1.5
1
0.5
0
Debugger
Breakpoints
Pause on Errors
Pause on Warnings
Pause on NaN or Inf
Pause on Output
Function call stack
Not paused in debugger
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